Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If A and B are two matrices such that $AB = B$ and $BA = A$, then $A^2 + B^2 =$

(A) $2AB$

(B) $2BA$

(C) $A + B$

(D) AB

Q.2

If $\mathrm{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \mathrm{a} & \mathrm{b} & -1 \end{pmatrix}$ then

(A) $\mathrm{A}^5 = \mathrm{A}$

(B) $\mathrm{A}^6 = \mathrm{A}$

(C) $\mathrm{A}^4 = \mathrm{A}$

(D) None of these

Q.3

Let $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$. The A is

(A) Nilpotent

(B) Idempotent

(C) Scalar

(D) None of these

Q.4

If A, B are square matrics, then $\mathrm{ABB'A'}$ is

(A) Symmetric

(B) Skew-symmetric

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(C) Data insufficient

(D) None of these

Q.5

If A is a skew symmetric matrix and $(\mathrm{I}+\mathrm{A})$ is non singular matrix then $\mathrm{B}=(\mathrm{I}-\mathrm{A})(\mathrm{I}+\mathrm{A})^{-1}$ is

(A) orthogonal matrix

(B) symmetric matrix

(C) skew symmetric matrix

(D) none of these

Q.6

Let $A=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$ and $B=\begin{pmatrix}a & 0\\ 0 & b\end{pmatrix}, a, b \in N$. Then

(A) there exist more than one but finite number of B 's such that $AB=BA$

(B) there exists exactly one B such that $AB=BA$

(C) there exist infinitely many B 's such that $AB=BA$

(D) there cannot exist any B such that $AB=BA$

Q.7

If $A=\begin{bmatrix}4 & 2\\ -1 & 1\end{bmatrix}$ and I is the identity matrix of order 2, then $(A-2I)(A-3I)$ is equal to

(A) I

(B) O

(C) $\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}$

(D) $\begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}$

Q.8

Let $F(\alpha)=\begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}$, where $\alpha \in R$. Then $[F(\alpha)]^{-1}$ is equal to

(A) $F(-\alpha)$

(B) $F\left(\alpha^{-1}\right)$

(C) $F(2\alpha)$

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(D) none of these

Q.9

If A is a matrix such that $A^2 - A + I = O$, then the inverse of A is

(A) $A - I$

(B) $I - A$

(C) $A + I$

(D) A

Q.10

The inverse of a skew symmetric matrix (if it exists) is

(A) a symmetric matrix

(B) a skew symmetric matrix

(C) a diagonal matrix

(D) none of these

Q.11

If $\begin{pmatrix} 1 & -\tan\theta \\ \tan\theta & 1 \end{pmatrix}\begin{pmatrix} 1 & \tan\theta \\ -\tan\theta & 1 \end{pmatrix}^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, then

(A) $a = b = 1$

(B) $a = \cos 2\theta, b = \sin 2\theta$

(C) $a = \sin 2\theta, b = \cos 2\theta$

(D) $a = 1, b = \sin 2\theta$

Q.12

If $P = \begin{bmatrix} \cos(\pi/6) & \sin(\pi/6) \\ -\sin(\pi/6) & \cos(\pi/6) \end{bmatrix}, A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP'$ then $P'Q^{2007}P$ is equal to

(A) $\begin{bmatrix} 1 & \sqrt{3}/2 \\ 0 & 2007 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 2007 \\ 0 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} \sqrt{3}/2 & 2007 \\ 0 & 1 \end{bmatrix}$

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(D) $\begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1 & 2007 \end{bmatrix}$

Q.13

If A and B are symmetric matrices, then $AB - BA$ is a

(A) symmetric matrix

(B) skew symmetric matrix

(C) diagonal matrix

(D) null matrix

Q.14

If A is a skew symmetric matrix, then A^4 is

(A) symmetric

(B) skew-symmetric

(C) invertible

(D) none of these

Q.15

The matrix product $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \end{bmatrix}$

(A) is skew symmetric symmetric

(B) invertible

(C) is orthogonal

(D) not invertible

Q.16

If $A = [a_{ij}]$ is a scalar matrix of order $n \times n$ such that $a_{ii} = k$ for all i, then $|A|$ is equal to

(A) nk

(B) $n + k$

(C) n^k

(D) k^n

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Q.17

If $AB = A$ and $BA = B$ and I is a unit matrix, then A^3 is equal to

(A) A

(B) B

(C) I

(D) none of these

Q.18

If $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ and $A(\text{adj}(A)) = \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then λ is equal to

(A) 1

(B) 2

(C) 3

(D) $\frac{1}{2}$

Q.19

If A is an orthogonal matrix and B is an orthogonal matrix, of the order same as that of A, then $C = A^{-1}BA$ is

(A) an orthogonal matrix

(B) symmetric matrix

(C) skew-symmetric matrix

(D) none of these

Q.20

Suppose A is a 3×3 skew-symmetric matrix. Let $B = (I + A)^{-1}(I - A)$. Then

(A) B is orthogonal

(B) B is skew symmetric

(C) $B^2 = O$

(D) B is a diagonal matrix

Q.21

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$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ -1 & -2 & -3 \end{bmatrix}$, then A is a nilpotent matrix of index

Q.22

The number of 3×3 matrices A whose entries are either 0 or 1 and for which the system of equations $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$ has exactly five distinct solutions is

Q.23

Suppose matrix A satisfies the equation $A^2 - 5A + 7I = O$. If $A^8 = aA + bI$, then value of a is

Q.24

Let $A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$, where $a > 0$. Sum of the series $S = \text{trace}(A) + \text{trace}\left(\frac{1}{2}A\right) + \text{trace}\left(\frac{1}{2^2}A^2\right) + \text{trace}\left(\frac{1}{2^3}A^3\right) + \ldots$ is

Q.25

Let $A = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{pmatrix}$ and $(10)B = \begin{pmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & 2 & 3 \end{pmatrix}$ If B is the inverse of matrix A, then α equals

Q.26

The least number of 3×3 non-singular matrices, with four entries as 1 and all other entries as 0 is

Q.27

Let $\omega \neq 1$ be a cube root of unity and S be the set of all non-singular matrices of the form $A = \begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$ where a, b, c are either ω or ω^2. Then number of distinct matrices in the set S is

Q.28

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Let $a, b, c \in \mathbf{R}$ be such that $a+b+c>0$ and $abc=2$. Let $A=\begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$ If $A^2=I$, then value of $a^3+b^3+c^3$ is

Q.29

If $P=\begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is the adjoint of a 3×3 matrix A and $|A|=4$, then α is equal to

Q.30

Let $A=\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B=\begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$. If $10A^{10}+\text{Adj}\left(A^{10}\right)=B$, then $b_1+b_2+b_3+b_4$ is equal to:

Answers & Solutions

Q.1 Answer:

$A+B$

Solution:

We have $AB=B$ and $BA=A$. Therefore,

$$A^2+B^2=AA+BB=A(BA)+B(AB)=(AB)A+(BA)B$$
$$=BA+AB=A+B \quad [AB=B \text{ and } BA=A]$$

Q.2 Answer:

$\mathrm{A}^5=\mathrm{A}$

Solution:

Here $\mathrm{A}^2=\mathrm{I}$, so we have

$$A^n=\begin{cases} A & \text{if } n \text{ is odd} \\ I & \text{if } n \text{ is even} \end{cases}$$

Q.3 Answer:

Nilpotent

Solution:

$$A^2=\begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix} \neq A$$

Hence not Idempotent

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$$A^3 = A^2A = \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$$
$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$$

Here A is nilpotent matrix of Index 3.

Q.4 Answer:

Symmetric

Solution:

$$\left(\mathrm{ABB'A'}\right)' = \left(\mathrm{B'A'}\right)'\left(\mathrm{AB}\right)'$$
$$= \left(\mathrm{AB}\right)\cdot\left(\mathrm{B'A'}\right) = \mathrm{ABB'A'}$$

So given matrix is symmetric

Q.5 Answer:

orthogonal matrix

Solution:

$$B' = \left((I-A)(I+A)^{-1}\right)' = \left((I+A)^{-1}\right)'(I-A)'$$
$$= \left((I+A)'\right)^{-1}(I-A)'$$
$$= (I-A)^{-1}(I+A)$$

obviously $\mathrm{B}\cdot\mathrm{B'} = \mathrm{I}$

Q.6 Answer:

there exist infinitely many B 's such that $AB = BA$

Solution:

$$AB = \begin{bmatrix} a & 2b \\ 3a & 4b \end{bmatrix}; BA = \begin{bmatrix} a & 2a \\ 3b & 4b \end{bmatrix}$$
$$AB = BA \Rightarrow a = b$$

Hence infinite solutions

Q.7 Answer:

O

Solution:

The expansion is clearly the characteristic equation i.e. $A^2 - \mathrm{Tr}(A)A + \mathrm{Det}(A)I = 0$

Q.8 Answer:

$F(-\alpha)$

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Solution:

We have,

$$F(\alpha)F(-\alpha) = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore F(-\alpha) = [F(\alpha)]^{-1}$$

Q.9 Answer:

$I - A$

Solution:

Pre-multiplying given equation by the inverse of A we get:

$A - I + A^{-1} = O$

Q.10 Answer:

a skew symmetric matrix

Solution:

We have, $A' = -A$

Now, $AA^{-1} = A^{-1}A = I_n \Rightarrow \left(AA^{-1}\right)' = \left(A^{-1}A\right)' = (I_n)'$

$\Rightarrow \quad \left(A^{-1}\right)A' = A'\left(A^{-1}\right)' = I_n \quad \Rightarrow \quad \left(A^{-1}\right)'(-A) = (-A)\left(A^{-1}\right)' = I_n$

Thus, $\left(A^{-1}\right)' = -\left(A^{-1}\right)$ **[inverse of a matrix is unique]**

Q.11 Answer:

$a = \cos 2\theta, b = \sin 2\theta$

Solution:

$$\begin{bmatrix} 1 & \tan\theta \\ -\tan\theta & 1 \end{bmatrix}^{-1} = \frac{1}{1+\tan^2\theta}\begin{bmatrix} 1 & -\tan\theta \\ \tan\theta & 1 \end{bmatrix}$$

$$\therefore \quad \begin{bmatrix} a & -b \\ b & a \end{bmatrix} = \cos^2\theta \begin{bmatrix} 1 & -\tan\theta \\ \tan\theta & 1 \end{bmatrix}\begin{bmatrix} 1 & -\tan\theta \\ \tan\theta & 1 \end{bmatrix}$$

$$\therefore \cos^2\theta \begin{bmatrix} 1-\tan^2\theta & -2\tan\theta \\ 2\tan\theta & 1-\tan^2\theta \end{bmatrix} = \begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}$$

$$\therefore \quad a = \cos 2\theta, b = \sin 2\theta$$

Q.12 Answer:

$$\begin{bmatrix} 1 & 2007 \\ 0 & 1 \end{bmatrix}$$

Solution:

Note that $P' = P^{-1}$.

Now $Q = PAP' = PAP^{-1} \Rightarrow Q^{2007} = PA^{2007}P^{-1}$

$\therefore P'Q^{2007}P = P^{-1}\left(PA^{2007}P^{-1}\right)P = A^{2007} = (I+B)^{2007}$

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$B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.

As $B^2 = 0$, we get $B^r = 0 \forall r \geq 2$.

Thus, by binomial theorem $A^{2007} = I + 2007B = \begin{pmatrix} 1 & 2007 \\ 0 & 1 \end{pmatrix}$

Q.13 Answer:

skew symmetric matrix

Solution:

Given $A' = A, B' = B$

Now $(AB - BA)' = (AB)' - (BA)' = B'A' - A'B' = BA - AB = -(AB - BA)$

i.e., $(AB - BA)' = -(AB - BA)$

Hence $AB - BA$ is a skew symmetric matrix.

Q.14 Answer:

symmetric

Solution:

We have, $A' = -A$

Now, $\left(A^4\right)' = (AAAA)' = A'A'A'A' = (-A)(-A)(-A)(-A) = A^4$

$\therefore$ **A^4 is symmetric matrix**

Q.15 Answer:

not invertible

Solution:

We have $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ -1 & -2 & 1 \end{bmatrix} = A$

$|A| = 0$,

We have $|A| = 0$, and hence A is not invertible.

Q.16 Answer:

k^n

Solution:

We have, $A = kI_n$

$\therefore \quad |A| = |kI_n| = k^n |I_n| = k^n$

Q.17 Answer:

A

Solution:

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We have $AB = A \Rightarrow ABA = AA = A^2$

$\Rightarrow \quad AB = A^2 \Rightarrow A = A^2 \Rightarrow A^3 = A \cdot A^2 = A^2 = A$

Q.18 Answer:

1

Solution:

$$A(\text{adj}(A)) = |A| \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \lambda = |A|$$

Now $|A| = \begin{vmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{vmatrix} = \cos^2\theta + \sin^2\theta = 1$

Q.19 Answer:

symmetric matrix

Solution:

$$C = A^{-1}BA$$
$$C^\top = \left(A^{-1}BA\right)^\top$$
$$C^\top = A^\top \left(A^{-1}B\right)^\top$$
$$C^\top = A^\top B^\top \left(A^{-1}\right)^\top$$
$$C^\top = A^\top B^{-1} \left(A^\top\right)^{-1}$$
$$C^\top = A^\top \left(A^\top B\right)^{-1}$$
$$C^\top = A^{-1}BA$$
$$C^\top = C$$

$\therefore$ It is symmetric

Q.20 Answer:

B is orthogonal

Solution:

We have

$$BB' = (I+A)^{-1}(I-A)\left[(I+A)^{-1}(I-A)\right]'$$
$$= (I+A)^{-1}(I-A)(I-A)'\left[(I+A)'\right]^{-1}$$
$$= (I+A)^{-1}(I-A)(I+A)(I-A)^{-1}$$
$$[\because \quad A' = -A]$$
$$= (I+A)^{-1}(I+A)(I-A)(I-A)^{-1}$$

[$I+A$ and $I-A$ commute]

$$= (I)(I) = I$$

Thus, B is orthogonal.

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Q.21 Answer:

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Solution:

$$A^2 = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ -1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ -1 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ A **is a nilpotent matrix of index 2**

Q.22 Answer:

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Solution:

We know that the system of equation

$AX = B$

has either (i) unique solution

or (ii) infinite number of solutions

or (iii) no solution.

Hence, there cannot exist any matrix A **such that** $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$

has five distinct solutions.

Q.23 Answer:

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Solution:

$A^2 = 5A - 7I \Rightarrow A^4 = \left(A^2\right)^2 = (5A - 7I)(5A - 7I)$

$= 25A^2 - 70A + 49I$

$= 25(5A - 7I) - 70A + 49I$

$= 55A - 126I$

$\Rightarrow A^8 = \left(A^4\right)^2 = (55A - 126I)(55A - 126I)$

$= 3025A^2 - 13860A + 15876I$

$= 3025(5A - 7I) - 13860A + 158766I$

$= 1265A - 5299I$

Thus, $a = 1265$.

Q.24 Answer:

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Solution:

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$A = I + B$, where $B = \begin{bmatrix} 0 & a \\ 0 & 0 \end{bmatrix}$

As $B^2 = O$, we get $B^r = O \quad \forall r \geq 2$.

$$\text{Thus, } A^r = (I+B)^r = I + rB$$

$$= \begin{bmatrix} 1 & ra \\ 0 & 1 \end{bmatrix} \forall r \geq 1.$$

$$\Rightarrow \text{trace}\left(\frac{1}{2^r}A^r\right) = \frac{1}{2^r}(2) = \frac{1}{2^{r-1}} \forall r \geq 1.$$

$$\therefore S = 2 + 1 + \frac{1}{2^2} + \frac{1}{2^3} + \cdots = \frac{2}{1 - \frac{1}{2}} = 4$$

Q.25 Answer:

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Solution:

$$10I = 10(AB) = A(10B)$$

$$\Rightarrow \begin{pmatrix} 10 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 10 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 10 & 0 & 5-\alpha \\ 0 & 10 & \alpha - 5 \\ 0 & 0 & 5+\alpha \end{pmatrix}$$

$$\Rightarrow \alpha = 5$$

Q.26 Answer:

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Solution:

The matrix

$$\begin{bmatrix} 1 & a & b \\ c & 1 & d \\ e & f & 1 \end{bmatrix}$$

where exactly one of a, b, c, d, e, f is 1 and rest of them are zeros, is invertible. There are six such matrices. Also, the matrix $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ **is invertible.**

Thus, there are at least 7 such matrices which are invertible.

Q.27 Answer:

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Solution:

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We have

$$|A| = 1\begin{vmatrix} 1 & c \\ \omega & 1 \end{vmatrix} - a\begin{vmatrix} \omega & c \\ \omega^2 & 1 \end{vmatrix} + b\begin{vmatrix} \omega & 1 \\ \omega^2 & \omega \end{vmatrix}$$

$$= 1 - c\omega - a\omega + ac\omega^2 = (1 - a\omega)(1 - c\omega)$$

A is non-singular $\Leftrightarrow |A| \neq 0 \Leftrightarrow a = c = \omega$.

Note that b can take any value ω or ω^2.

Thus, there are two non-singular matrices.

Q.28 Answer:

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Solution:

$$A^2 = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}\begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix} = \begin{bmatrix} \alpha & \beta & \beta \\ \beta & \alpha & \beta \\ \beta & \beta & \alpha \end{bmatrix}$$

where $\alpha = a^2 + b^2 + c^2$,

$\beta = bc + ca + ab$.

As $A^2 = I$**, we get**

$a^2 + b^2 + c^2 = \alpha = 1$

$bc + ca + ab = 0$

Now, $(a + b + c)^2 = \alpha + 2\beta = 1$

$\Rightarrow a + b + c = 1$

We have

$a^3 + b^3 + c^3 - 3abc$

$= (a + b + c)\left(a^2 + b^2 + c^2 - bc - ca - ab\right)$

$= 1$

$\Rightarrow a^3 + b^3 + c^3 = 7$

Q.29 Answer:

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Solution:

$|P| = |A|^2 = 16$

$\Rightarrow (1)(0) - \alpha(4 - 6) + (3)(4 - 6) = 16$

$\Rightarrow 2\alpha - 6 = 16 \Rightarrow \alpha = 11$

Q.30 Answer:

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Solution:

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Let $A = I + C$ **where** $C = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

$\text{AS}\, C^2 = O$, **we get** $C^r = 0 \forall r > 2$

Now,

$A^{10} = (I + C)^{10}$

$= I + 10C + {}^{10}C_2 \left(C^2\right) + \ldots + {}^{10}C_{10} \left(C^{10}\right)$

$= I + 10C + O = \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow \text{Adj}\left(A^{10}\right) = \begin{bmatrix} 1 & -10 \\ 0 & 1 \end{bmatrix}$

$10\ \text{A}^{10} + \text{adj}\left(\text{A}^{10}\right) = 10 \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -10 \\ 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 10 & 100 \\ 0 & 10 \end{bmatrix} + \begin{bmatrix} 1 & -10 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 90 \\ 0 & 11 \end{bmatrix}$

$10\ \text{A}^{10} + \text{adj}\ \text{A}^{10} = \begin{bmatrix} 11 & 90 \\ 0 & 11 \end{bmatrix} = \text{B} = \begin{bmatrix} \text{b}_1 & \text{b}_2 \\ \text{b}_3 & \text{b}_4 \end{bmatrix}$

$\text{b}_1 + \text{b}_2 + \text{b}_3 + \text{b}_4 = 11 + 90 + 0 + 11 = 112$

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